

MOHD UMAR HASHMI

AI Integration Engineer | Android Developer | Algorithmic Trading Systems Engineer | Product Builder

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<https://umarhashmi-7.github.io/umar-portfolio/>

PROFESSIONAL SUMMARY

Results-driven Software & AI Integration Engineer specializing in Python backend engineering, distributed systems, and AI/LLM product development, with additional expertise in Android development and quantitative analysis. Architected two production-grade algorithmic trading platforms: the UMAR SMC Bot — an institutional-grade automated trading system validated across 1,000+ signals on a 3-year historical backtesting framework with a 52.06% win rate and 1.98 profit factor (historical backtest; live demo account validation ongoing) — and the V7.2 Automated Multi-Asset MT5 Trading Platform, a multi-process backend system built around the MetaTrader5 API with real-time monitoring, watchdog recovery, and remote operational control. Independently built SIUFIT — a 10+ module production Android fitness ecosystem (Kotlin, Firebase, Groq/LLaMA 3.3-70B, ML Kit). Brings a rare blend of deep backend execution, fault-tolerant distributed architecture design, and measurable impact: 30%+ academic improvement for 20+ students, AI/ML workshop facilitation for 30 educators, and 800+ participant event management. BCA Coursework Completed | Degree Award Pending (2026).

CAREER HIGHLIGHTS

Two Production Algorithmic Trading Platforms | 52.06% Win Rate | 1.98 Profit Factor* | 10+ Module Production Android App
30%+ Academic Score Improvement | AI/ML Workshop — 30 Educators | 800+ Participant Event Management
100% Client Retention (3+ Years) | NPTEL Elite Cert — IIT Madras 2025 | *Historical Backtest Results

CORE COMPETENCIES

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| ✓ Python Backend Development | ✓ MVVM / MVC Architecture |
| ✓ Distributed Systems & Concurrency | ✓ Production Deployment & Validation |
| ✓ AI & LLM API Integration | ✓ Offline-First & SQL Design |
| ✓ Automation Engineering | ✓ System Design & Performance Optimization |
| ✓ Firebase & Cloud Architecture | ✓ Monitoring & Fault Tolerance |
| ✓ REST API Design & Consumption | ✓ Multi-processing & Process Synchronization |
| ✓ Backend Architecture & Long-running Services | ✓ Structured Logging |
| ✓ Event-Driven Architecture | ✓ SEO & Digital Marketing |
| ✓ Public Speaking & Mentoring | ✓ Cross-functional Collaboration |

TECHNICAL SKILLS

Mobile Development: Android Studio · Kotlin · Java · XML Layouts · Android SDK · Jetpack Components · CameraX · ML Kit

Python / Backend & Algorithmic Systems: Python · Pandas · NumPy · MetaTrader5 API · Algorithm Design · Backtesting Frameworks · Trading Automation · Flask · Dash · Plotly / Plotly.js · AsyncIO · Multiprocessing · Concurrency · Distributed Systems · Automation Engineering · Monitoring Systems · Watchdog Architecture · Financial API Integration · REST APIs · Data Processing

Backend & Database: Firebase Auth · Firestore · Firebase RTDB · SQLite · Room Database · Google Sign-In · FileLock · Windows Process Management

AI / ML: Groq API · LLaMA 3.3-70B · ML Kit · Prompt Engineering · LLM Integration · AI Workflow Automation

Programming: Python · Kotlin · Java · C · C++ · HTML · CSS · JavaScript · OOP Principles · Data Structures

APIs & Integration: USDA FoodData API · Open Food Facts API · Delta Exchange API · Telegram Bot API · REST Architecture · Retrofit · JSON Parsing

Tools & DevOps: Git · GitHub · Linux · VS Code · Android Studio · MS Office (Advanced Excel) · Figma · Google Workspace

Data & Analytics: Excel Data Analysis · Pivot Tables · VLOOKUP · KPI Tracking · Data Visualization · Reporting Dashboards

Business & Marketing: Requirements Analysis · Stakeholder Communication · SEO Strategy · Content Development · Digital Marketing

KEY PROJECTS

UMAR SMC Bot — Institutional-Grade Algorithmic Trading System

Python · Pandas · NumPy · Plotly.js · REST APIs · Delta Exchange API · HTML · CSS · JavaScript

- ▶ Architected and developed an end-to-end institutional-grade Smart Money Concepts (SMC) and ICT-based algorithmic trading system in Python, covering the full software engineering lifecycle: architecture design, development, testing, debugging, validation, and production deployment preparation.
- ▶ Designed automated market structure analysis engine detecting Swing Highs, Swing Lows, Break of Structure (BOS), and Change of Character (CHOCH) reversals using lookback-3 pivot algorithms with EMA20/EMA50 dual-filter trend confirmation — preventing future data leakage and false breakout signals while ensuring mathematical consistency across historical and live environments.
- ▶ Engineered a 3-year historical backtesting framework (June 2023–June 2026) validating 1,020 trading signals; achieved historical backtest metrics: 52.06% win rate, 1.98 profit factor, 10.17% max drawdown, and Sharpe/Sortino ratios of 5.15/12.64. [Historical Backtest Results — Live Demo Account Validation Ongoing]
- ▶ Built a real-time analytics dashboard (Python, HTML, CSS, JavaScript, Plotly.js) with micro-state event dispatching architecture, and implemented production-grade safety systems — 1-second Emergency Kill Switch, Liquidity Protection Logic, Metadata Persistence Layer for post-reboot recovery, and Auto-Reconnect Mechanisms — integrated with OCO bracket execution via the Delta Exchange Demo REST API.
- ▶ Achieved 100% signal parity between the historical backtesting engine and live scanner validation across 1,000+ evaluated bars, proving algorithmic correctness for production deployment on a live demo account.

V7.2 Automated Multi-Asset MT5 Trading Platform

Python · MetaTrader 5 API · Pandas · NumPy · Flask · Dash · Plotly · Telegram Bot API · AsyncIO · Multiprocessing · FileLock · JSON · Windows Process Management

- ▶ Architected a production-grade, multi-process automated trading platform integrating the official MetaTrader5 API for real-time market data ingestion, order execution, and position management, with independent strategy daemons executing simultaneously under a centralized supervision layer.
- ▶ Designed a modular execution engine separating strategy logic, centralized trade execution, and risk management — implementing position sizing, exposure limits, and drawdown protection as decoupled, independently testable services.
- ▶ Built a Flask + Dash + Plotly monitoring dashboard delivering real-time visibility into system health, process status, and platform performance across all running strategy daemons.
- ▶ Integrated a Telegram Bot API console for remote operational control — enabling start/stop of individual strategies, live status checks, and automated alerting without direct server access.
- ▶ Engineered a watchdog recovery system with automated process detection and restart logic, achieving stable 24/7 long-running service operation by recovering crashed strategy daemons without manual intervention.
- ▶ Implemented FileLock-based synchronization to prevent race conditions across shared state files, paired with JSON-based state persistence enabling clean position and process recovery after crashes or restarts.
- ▶ Developed thread-safe, timestamped concurrent logging across all processes and validated candle-by-candle parity between the backtesting engine and live execution layer, profiling and tuning CPU/memory usage for sustained multi-process workloads.

SIUFIT — Production Android Fitness & Nutrition Ecosystem

Kotlin · Firebase · Groq/LLaMA 3.3-70B · ML Kit · USDA API · SQLite · MVVM

- ▶ Identified a gap in personalized AI-powered fitness tracking for Indian users and independently architected SIUFIT — a 10+ module production Android application (MVVM) solving nutrition logging, workout planning, hydration tracking, and AI-guided coaching in a unified platform.
- ▶ Integrated Groq/LLaMA 3.3-70B AI chatbot for real-time contextual fitness guidance; connected ML Kit barcode scanner with USDA FoodData Central API (Open Food Facts fallback) for accurate live nutritional data across 300,000+ food entries.
- ▶ Implemented offline-first architecture using SQLite local storage with Firebase Firestore bidirectional sync — ensuring full feature availability without internet connectivity and seamless state restoration on reconnection.
- ▶ Engineered BMR/TDEE metabolic calculators and meal-wise macronutrient logging with sub-100ms UI response across all 10+ modules; delivered a dark glass UI theme with multi-tier onboarding to reduce new-user drop-off and improve cross-module feature discoverability.

Android Application Suite — 4 Production Apps

Kotlin · Java · Firebase Auth · SQLite · Room · MVC/MVVM

- ▶ Independently built and shipped 4 production-ready Android applications — Task Manager, Notes App, Firebase Authentication System, and College Utility Tools — applying MVVM/MVC architecture, SOLID principles, and modular component design throughout.
- ▶ Achieved 40%+ reduction in app load times via modular reusable architecture with standardized shared libraries; reduced per-feature development cycle time by centralizing authentication and data access layers across all 4 applications.
- ▶ Integrated SQLite offline persistence with Firebase Authentication for secure cross-device access — establishing consistent data integrity and session management patterns applicable to enterprise-grade Android projects.

Inventory & Billing Management System

Python · SQLite · OOP · Database Design

- ▶ Developed a Python-based Inventory & Billing Management System for S.S. Pharmaceuticals covering inventory tracking, stock management, automated invoice generation, and billing automation — improving operational accuracy and efficiency across the business.
- ▶ Designed and managed database-driven solutions for product records, inventory levels, transaction history, and sales data; collaborated with business stakeholders to gather requirements, analyze workflows, and deliver aligned software solutions.

PROFESSIONAL EXPERIENCE

Freelance Instructional Technologist & Curriculum Designer

Self-Employed

Aug 2022 – Present

Moradabad, Uttar Pradesh

- ▶ Designed and delivered personalized, syllabus-aligned Computer Science and Mathematics curricula for 20+ students, consistently achieving 30%+ improvement in standardized assessment scores through gap-targeted, differentiated instruction strategies.
- ▶ Engineered adaptive learning frameworks tailored to individual learner profiles, systematically reducing concept-to-mastery timelines by identifying cognitive gaps and applying targeted content intervention strategies.
- ▶ Sustained 100% client retention across 3+ years of continuous engagement through structured progress tracking, transparent communication, and documented measurable learning outcomes — demonstrating strong entrepreneurial client management capability.
- ▶ Developed reusable instructional materials — concept notes, practice assessments, and topic-wise learning progressions — building a foundational LMS-ready content library applicable to digital EdTech platforms.

Inventory & Billing System Developer (Python)

S.S. Pharmaceuticals

May 2024 – Jul 2024

Moradabad, Uttar Pradesh

- ▶ Developed, tested, debugged, and deployed a Python-based Inventory & Billing Management System — delivering inventory tracking, stock management, invoice generation, and billing automation aligned to pharmaceutical business operations.
- ▶ Collaborated with business stakeholders to gather requirements, analyze operational workflows, and implement database-driven software solutions improving data accuracy, operational efficiency, and reporting capabilities.

LEADERSHIP & COMMUNITY INVOLVEMENT

Event Organizer & Operations Coordinator

Teerthanker Mahaveer University

2023 – Present

Moradabad

- ▶ Orchestrated 5+ large-scale university events — sports competitions, fresher orientation, farewell celebrations — independently managing end-to-end logistics, vendor negotiations, AV/technical infrastructure, and team coordination for 800+ participants per event.
- ▶ Selected by university administration to deliver opening and closing addresses at flagship events to audiences of 800+; recognized for structured, authoritative public communication.

AI/ML/Deep Learning Workshop Facilitator

RAN Public School

2024

Moradabad

- ▶ Designed and delivered a full-day hands-on AI/ML/Deep Learning workshop curriculum for 30 school educators — translating graduate-level AI concepts into curriculum-relevant practical exercises; received 'highly effective' instructor rating from institutional feedback.

Communication Skills Mentor

IEEE Student Branch SB STB15123, TMU

2024

Moradabad

- ▶ Coached 15+ emerging student leaders in professional communication, public speaking, and presentation design through structured peer-mentorship sessions; equipped participants with technical storytelling frameworks applied in IEEE events and intercollegiate competitions.

College Sports Representative

Teerthanker Mahaveer University

2023 – Present

Moradabad

- ▶ Represented TMU in intercollegiate football and multi-sport competitions, demonstrating consistent athletic discipline, high-pressure team coordination, and competitive resilience.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Advanced / Industry

- ▶ NPTEL Elite Certification — Social Networks — IIT Madras (2025)
- ▶ AI for Business Professionals — HP LIFE (2026)
- ▶ IT for Business Success — HP LIFE (2026)
- ▶ AI Tools & ChatGPT Workshop — be10x (2025)

Professional Development

- ▶ Communication Mentor Certification — IEEE SB STB15123, TMU (2024)
- ▶ Prompt Engineering & AI Applications — Mind Labs (2023)
- ▶ Artificial Intelligence Fundamentals — Mind Labs (2023)
- ▶ Digital Marketing Fundamentals — Google (2023)
- ▶ Content Writing & E-Book Writing — Mind Labs (2022)
- ▶ Excel Data Analysis — Coursera (2021)
- ▶ HTML/CSS Fundamentals — Open Weaver (2021)
- ▶ IBM Skills Network — Big Data for Beginner – 3C (2026) | Big Data · Hadoop · Apache Pig · MapReduce · YARN · Distributed Computing

EDUCATION

Bachelor of Computer Applications (BCA)

Teerthanker Mahaveer University, Moradabad, UP | Coursework Completed | Degree Award Pending

Expected 2026

Senior Secondary — Class 12 (CBSE)

Bonne Anne Public School, Moradabad

2023

Secondary — Class 10 (CBSE)

Bonne Anne Public School, Moradabad

2021

ADDITIONAL INFORMATION

Languages: English (Professional Fluency) · Hindi (Native) | Availability: Immediate joining · Open to full-time, remote, and contractual roles | Sports: University-level football representative; intercollegiate multi-sport events (TMU)

Areas of Interest: Backend Systems & Distributed Architecture · Algorithmic Trading & FinTech Engineering · Android & Cross-Platform Mobile Development · AI Engineering & LLM Product Integration · Automation & Production Infrastructure · Quantitative Analysis & Financial Technology · NLP & LLMs · Data-Driven Decision Making